2
Concepts, Definitions and Composition

Serkan Arslanalp, Wolfgang Bergthaler, Philip Stokoe, and Alexander F. Tieman

1. Introduction

In February 1989, workers in New York were putting the finishing touches to a very special clock. The brainchild of New York real estate developer Seymour Durst, the 11-by-26-foot digital clock cost $100,000 and was erected one block from Times Square. The clock was the US National Debt Clock and since then, apart from a brief period between September 2000–July 2002 when debt was falling, the clock, and its replacement installed in 2004, has steadily tracked the rise in US debt, from $2.7 trillion when the clock was unveiled in February 1989, to $22.4 trillion at August 13, 2019.

According to Mr. Durst's son, his father had been obsessed with debt since the early 1980s. In 1980, during the holiday season he sent cards to members of the US congress which included the message "Happy new year. Your share of the national debt is $35,000." Mr. Durst's idea has inspired others, and a google search for "debt clock" quickly takes you to numerous websites with debt clocks for countries all over the world. There is just one problem with all these clocks, which is this. *What do they measure*?

Mr. Durst's clock tracks a concept of debt that the US Treasury calls "Total Public Debt Outstanding." This figure was $22.0 trillion at December 31, 2018. But there are other estimates of debt for the United States. The IMF World Economic Outlook (WEO) Database publishes an aggregate called "general government gross debt." The April 2018 database presents this number at the end of 2018 as $21.7 trillion—a difference of $270 billion (approximately 1.3 percent of GDP). The IMF also publishes data on government liabilities in the Government Finance Statistics (GFS) database, drawing on data reported

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to the IMF's Statistics Department by the US Bureau of Economic Analysis and Federal Reserve Board. This database shows total liabilities of the US general government of $28.4 trillion, $6.8 trillion higher than the WEO number, and $6.4 trillion higher than on the debt clock (almost 19 percent of GDP). The IMF's Fiscal Affairs Department (FAD) has also compiled data for total public sector liabilities for the United States (IMF 2018). This experimental data set estimates US public sector liabilities to be almost $36 trillion in 2016, a figure almost twice the number on the New York clock.

Finally, academics such as Larry Kotlikoff and some politicians have claimed the real number is higher still. Senator Ben Sasse, a Republican from Nebraska, made waves in an April 2017 town hall meeting in Elkhorn, Nebraska, by asserting that the real value of public debt was between $70–75 trillion—more than three times the value shown on the debt clock during 2017. The *Washington Post* factchecked this assertion and awarded Mr. Sasse "one Pinocchio" on their fact checking scale (which runs from one to four pinocchios), which they say could be viewed as being "mostly true." Kotlikoff was widely quoted in 2015 as saying the true debt figure for the United States then was $210 trillion, a figure which was arrived at through so called "generational accounting," which defines government debt as the net present value of future government cash flows—which if true would clearly mean Mr. Durst should have installed a significantly bigger clock.

The truth is, the value of debt is hugely dependent on three factors: the institutional coverage of the debt; the instrument coverage of the debt; and the valuation of the debt. The different measures of debt discussed above cover different parts of the American government or wider public sector, include different kinds of government liabilities and/or contingent liabilities, and use different valuations. Globally, different countries use different concepts of debt, making cross-country comparability of sovereign debt difficult.

In any discussion of sovereign debt, public debt, national debt, or government debt, the *definition* of debt matters enormously. What kind of liabilities are included, belonging to which entities, and how they are valued is the difference between debt for the United States being as little as $15 trillion or more than $70 trillion. Indeed, it might be more accurate to say that there isn't a single measure of sovereign debt, but a number of complementary measures that policymakers can use to understand their fiscal position and the sustainability (or not) of their fiscal position. Section 2 of this chapter discusses these key concepts of institutional coverage, valuation, and instrument coverage which are critical to any discussion of sovereign debt.

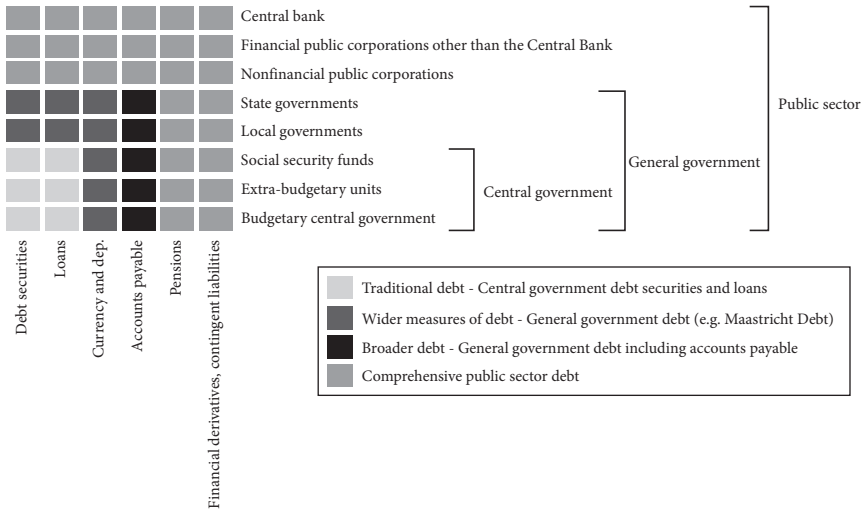

Figure 2.1 Debt in two dimensions

These concepts are illustrated in Figure 2.1, which shows the institutional coverage and instrument coverage dimensions of debt. Focusing on a narrow measure of debt, such as just the narrowly defined debts of the central or even general government may mean analysis overlooks significant liabilities and sources of fiscal risk for the government or in the wider public sector.

Section 3 looks specifically at the public sector as the broadest possible concept of coverage, building on IMF (2018). In doing so it recognizes public sector assets as well as liabilities. These assets may reside inside the general government, or in public corporations (often referred to as state-owned enterprises or SOEs), outside the standard government accounts but nevertheless controlled by the state. Taken together, these elements provide the most comprehensive view of public wealth, in the form of a country's public sector balance sheet (PSBS). In most countries, the PSBS is little understood, poorly measured, and only partly managed, with analysis instead focused on "standard" fiscal flows—revenues, expenditures, and deficits—and gross debt. This misses large swaths of government activity and can fall victim to illusory fiscal practices.

Finally, whatever the level of debt, the composition of the debt, in terms of the currency it is issued in, the maturity profile, and the nature of the creditor to whom debt is owed can be very important for debt management and sustainability. Section 4 of this chapter discusses the composition of debt from a debtor (supply) and creditor (demand) perspective.

2. Measures of Sovereign Debt

A. Institutional coverage of debt

This book is about "sovereign debt," but this is a slippery concept. Online sources have numerous definitions, including debt of the national government,[1] debt of the central government,[2] government debt in a currency other than a government's own national currency,[3] or debts owed or guaranteed by the national government.[4] Some consider sovereign debt to include just the liabilities of the central government, some extend this concept to also include the debts of the Central Bank. Finally, some people use the term *sovereign debt* interchangeably with the term public debt[5] and public debt interchangeably with the term government debt[6] but these may not refer to the same thing.

Assuming that sovereign debt is only the debt of government, what is the definition of *government*, given that the nature and structure of government differs significantly across countries? Consider the G7 economies; within this group are three countries with federal systems of government (Canada, Germany, and the United States; and four countries with more centralized systems (France, Italy, Japan, and the UK). Because of this, comparing the debt of the Federal Government of Canada, or Germany, or the United States with the debt of the Central Government of the other four G7 countries will be highly misleading. Instead, most statisticians would argue that debt should be compiled for the *general government*,[7] a broader concept that includes all government entities whether these are national governments with jurisdiction over the entire country; state, provincial, or regional governments (such as Ontario, Texas, or Western Australia); or local governments and municipalities. General government also includes a group of entities called social security funds, which administer social insurance schemes for pensions, health, and unemployment benefits.

[1] http://lexicon.ft.com/Term?term=sovereign-debt
[2] https://www.investopedia.com/terms/s/sovereign-debt.asp
[3] https://www.collinsdictionary.com/dictionary/english/sovereign-debt
[4] https://blogs.imf.org/2017/02/23/dealing-with-sovereign-debt-the-imf-perspective/
[5] https://www.thebalance.com/what-is-the-public-debt-3306294
[6] https://www.focus-economics.com/economic-indicator/public-debt
[7] This concept of general government is defined in statistical manuals like the UN System of National Accounts 2008, the IMF Government Finance Statistics Manual 2014, the IMF Public Sector Debt Statistics Guide for Compilers and Users 2011 and is used by the European Union in their regional statistical manual, the European System of Accounts 2010, which is the statistical underpinning for Maastricht Debt—defined as general government consolidated gross debt.

Government units, in the statistical manuals, are defined as legal entities established by political processes that have legislative, judicial, or executive authority over other entities within a given area with responsibility for the provision of public goods and services, to redistribute income and wealth; that engage primarily in non-market production; and that are financed primarily out of taxation, or other compulsory transfers. While this definition is relatively easy to apply for many government units, at the borderline, there are numerous grey areas. General government in many countries includes entities that a lay person might not consider to be part of government, such as public transport companies, publicly owned financial institutions (such as EXIM or Development Banks), or other state-owned enterprises.

In addition to government units, the manuals recognize the existence of a wider public sector. This includes entities under public control that do not meet the definition of a government unit. These entities are typically said to be engaged in market production, selling goods and services to households and businesses on a market basis. Often referred to colloquially as state-owned enterprises or public enterprises, statisticians refer to these publicly owned and/or controlled non-government entities as *public nonfinancial corporations* or *public financial corporations* (see Figure 2.2). They can take any form and operate in any industry. The size and complexity of the public sector varies considerably, across countries and over time. Section 3.C takes this broadest possible institutional coverage of public finances as its point of departure. In China and Russia, the public sector is extensive and covers many or almost

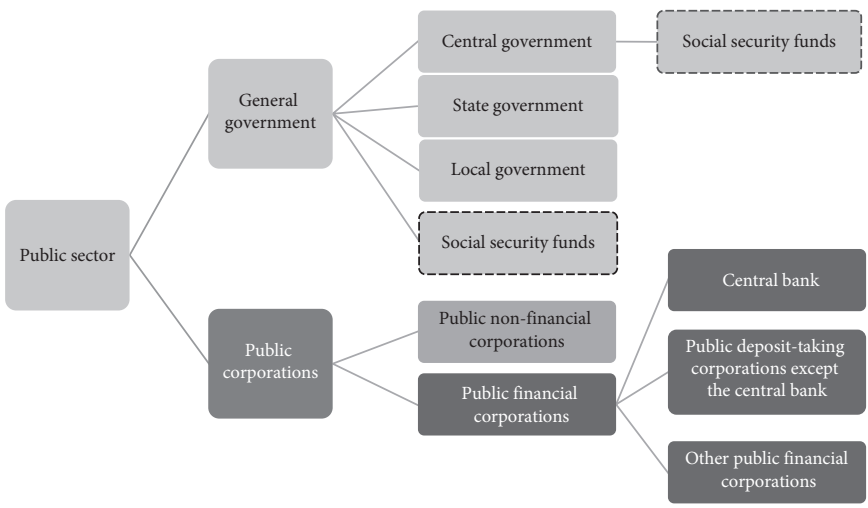

Figure 2.2 Public sector and its main components

all of the main parts of the economy. In other countries public ownership is much less extensive. In the UK the public sector in 2018 looks considerably different to the public sector in 1980, following waves of privatization (see Case Study 2.1) Entities are considered to be public units if they are controlled, rather than owned by the government (reflecting government's unique ability to intervene in an economy), but while this is often easy to recognize, at the

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Case Study 2.1 Gaming the perimeter—Network Rail (UK)

During the 1980s and 1990s, the UK Government privatized many formerly state-owned enterprises. Over the fifteen years from 1981 the UK sold off British Sugar, British Telecom, British Gas, British Airways and the British Airports Authority, British Steel, British Coal, Water, and Electricity companies, and many more.

In 1996 the government concluded the privatization of the UK's rail industry via a complex privatization whereby the old British Railways Board was split into many separate companies, including separate units responsible for managing the infrastructure, passenger rolling stock, six separate freight companies, six track renewal units, seven infrastructure maintenance units, and a number of other companies.

The infrastructure company was Railtrack PLC, which debuted on the London Stock Exchange in May 1996, but very quickly things began to go wrong. The new government ultimately pulled the plug on Railtrack and in October 2002 placed control over the infrastructure under a new company, a not-for-profit entity called Network Rail.

Network Rail was carefully established by the government to be classified outside of the public sector. Consequently, despite the extensive subsidies being provided to the firm, the UK Office of National Statistics (ONS) classified Network Rail as part of the private sector from shortly after its creation, from March 2003. Network Rail remained classified outside of the public sector for the whole of the 2000s, even as its debt steadily climbed.

By the end of 2007 Network Rail Ltd had amassed liabilities of £25bn, including £18.5bn of loans. At the time, UK public sector net debt stood at £535.7bn or 37.8 percent of GDP. The careful structuring of Network Rail had enabled the UK government to keep debt of more than 1 per cent of GDP off the books!

Continued

Case Study 2.1 *Continued*

By 2011, the classification of Network Rail was coming under strain. In January 2011 the European Union Statistical Authority (Eurostat) first raised the issue of the classification. In 2013, Eurostat raised the issue again, pointing to new statistical guidance. Meanwhile Network Rail's debt had continued to rise. By the end of 2012 its liabilities had climbed to over £39bn, including £29bn of loans—almost 2 percent of GDP. Faced with new guidance, and a skeptical Eurostat, the ONS reached a new conclusion in December 2013 and retrospectively reclassified Network Rail as part of the public sector back to 2004. Not only that, the company should be classified as part of general government.

With the stroke of a statistician's pen, UK public sector and UK general government debt was rewritten. By the time Network Rail was reclassified, the UK's debt had already ballooned as the effects of the financial crisis continued to be felt, such that the inclusion of Network Rail's debt, and even larger debts (such as those of registered social landlords) was a relatively small drop in the ocean, but had these classification decisions been applied during the mid-2000s, it is likely that UK rail policy would have been significantly different if "putting things on the Network Rail credit card" had also meant being recorded as government or public sector debt.

borderline things again can be tricky (for examples see, e.g., Mano and Stokoe 2017 or Box 1.3 in IMF 2018).

State-owned banks or other public financial corporations have particular implications for public sector debt, as they will invariably have large balance sheets, and thus large liabilities or debts. Public sector debt for countries with many publicly owned banks will look very different, and much higher, than countries with few or no publicly-owned banks. In Germany, public financial corporations including the publicly owned Landesbanken and Sparkassen have total liabilities of approximately 100 percent of GDP. France, with far less extensive public ownership of financial corporations, has public financial corporations with less than half of the German total.

The Central Bank presents a final, special case. By long standing convention, the National Central Bank is not classified inside the general government, but instead classified as a public financial corporation, even though in many ways it resembles a government unit. In some countries Central Banks are said to be "independent," in others, the Central Bank has much

less operational independence. However Central Banks can and do issue debt of their own and have considerable liabilities. Recognizing this realty, central banks' debt may be included in a sovereign's debt restructuring at the election of the sovereign (Hagan 2005). This has important implications for government or sovereign debt, and makes country comparisons difficult between countries where Central Banks issue their own bonds and debt instruments to manage liquidity, and the majority of countries whose Central Banks do not do this and instead use government issued instruments like Treasury Bills to manage liquidity.

In addition, the exclusion of the Central Bank from general government also has implications for sovereign debt if the Central Bank acts as a creditor to the government. While historically this has often been the case in developing or emerging market economies, in recent years it has also become a feature of advanced economies, especially in those countries that have engaged in quantitative easing (QE). Returning to the United States, after 10 years of QE, a significant part of the US public sector debt is now held by the US Federal Reserve (the Fed). In Q1 2018 the Fed held $2.4 trillion of Federal Government securities, 14 percent of the total stock. The Fed held a further $1.7 trillion of Fannie Mae and Freddie Mac debt securities, around a quarter of Fannie and Freddie's total debt. In the UK, the Bank of England Asset Purchase Facility (the entity established by the Bank of England as the vehicle for QE) holds UK government gilts worth £435 billion, this is around 20 percent of the total UK gilts owed by the UK government and means that a large part of the UK general government debt is owed to the wider public sector. The Bank of Japan, which has also engaged in significant QE over a long period, holds Japanese Government securities worth ¥457 trillion, this is almost half of the Japanese general government liabilities in the form of debt securities, which stood at ¥996 trillion at the end of 2016. This growth in Central Bank holdings of government debt, can be seen in Figure 2.3.

Institutional coverage of sovereign debt or national debt reported at the national level varies considerably, presenting challenges for cross country analysis (see Annex 2.A). In advanced economies, most countries will compile statistical measures of general government debt, in line with the statistical manuals. This includes in the European Union, where all countries are required to compile and report Maastricht Debt, which is general government consolidated gross debt in line with the European System of Accounts 2010. However, at the national level the headline national measures of debt are often something else. This is true in a number, but not all, federal countries. In the United States, the headline measure of debt, that features on Mr. Durst's clock, is

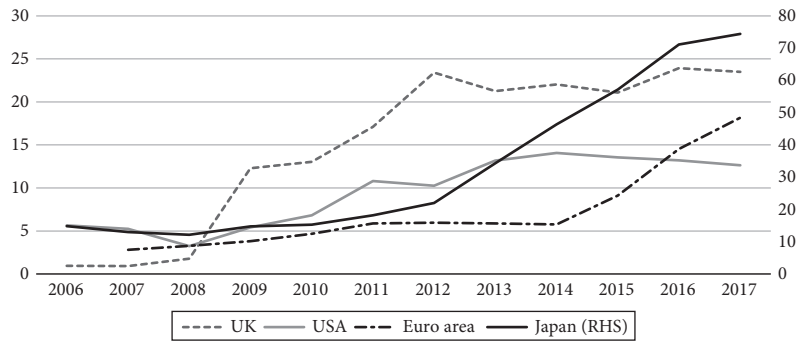

Figure 2.3 Central bank holdings of government debt, 2007–18 (% of GDP)
Source: IMF International Finance Statistics.

federal government debt only, the debts of state and local governments in the United States are excluded. In the UK, the headline debt measure is called Public Sector Net Debt,[8] and extends beyond the general government to include the liabilities of public corporations and the Bank of England and includes some offsetting liquid assets. In Latin America, measures of public debt often extend to include the debts of some or all state-owned enterprises, and sometimes the Central Bank. In many low-income or developing economies countries, the focus is on debt measures that include both the formal government debt and explicit government guaranteed debt of non-government units (but leave out debt without an explicit guarantee).

There is arguably no single correct coverage of debt. Proponents of general government debt measures draw a line between the predominately tax financed nature of government, compared to the market revenues of government owned corporations. However, for many countries the presence of explicit or implicit government guarantees means that the debt of public owned corporations can be a significant course of fiscal risk, should these guarantees be called—for those who draw a line between the public sector and the private sector, the key metric is total public sector debt. However, for users of debt data it is important to understand what the coverage is, to what extent data has been consolidated by eliminating intra-public sector holdings, and what might be lurking outside the reporting perimeter.

[8] During the global financial crisis, Public Sector Net Debt ballooned as the UK government nationalized several previously private large banking groups thus expanding debt. This prompted the UK authorities to create a new headline measure of debt, called "Public Sector Net Debt Ex" that excluded the temporary impacts of the financial crisis.

B. Valuation of debt

A second important factor in thinking about sovereign debt is the valuation of that debt, especially, but not only, in relation to government debt securities. Broadly speaking, debt can be valued in three main ways, but which valuation is used can have a significant impact on the value of debt stock.

For most countries, debt is valued at *face value*, or the value to be repaid at maturity. Some countries report debt at *nominal value*, defined as the principal sum borrowed plus interest accrued and not yet repaid. In National Accounts, the basic principle is to value debt at the *market value*—which is the value for which a bond will change hands in the secondary market (and which can be readily observed in the case of government bonds with well-developed capital markets, such as the bonds for most advanced economies. For debtors, the face and nominal values are critical, but for creditors, the market value of the bonds, which reflects the default risk is also important, as it reflects the likelihood of them receiving their money back.

At maturity of an instrument, all these valuation methods will arrive at the same ultimate point, but during the lifetime of the bond these different valuations can diverge significantly. In addition, valuation has a major impact depending on the nature of the bond that is issued.

Consider bonds issued at a discount, or even a deep discount. Examples of these include Treasury Bills, which typically have short-term maturities, but other longer term instruments issued at a discount are not unknown. These bonds may pay a small coupon, or no coupon at all. Instead investors pay a discounted amount at issuance, and then receive the face value of the instrument at maturity. The difference between the issue price and redemption amount is essentially equivalent to the interest. Now consider the difference between the face value and nominal value of such instruments. At issuance, the nominal value is lower than the face value, but over the life of the instrument, the nominal value increases as interest accrues before meeting face value at the time of maturity (Figure 2.4).

Depending on the local public debt law, and definitions in use for headline measures of debt, government debt management offices may have incentives to issue more, or less of, particular types of debt instrument. From an investor's perspective, as long as the government is deemed solvent, and as long as the eventual yield is satisfactory, because they will typically record the debt at nominal or market value in their financial statements, they will be indifferent to the types of instrument issued, but the ability of government debt managers to game the headline debt numbers through

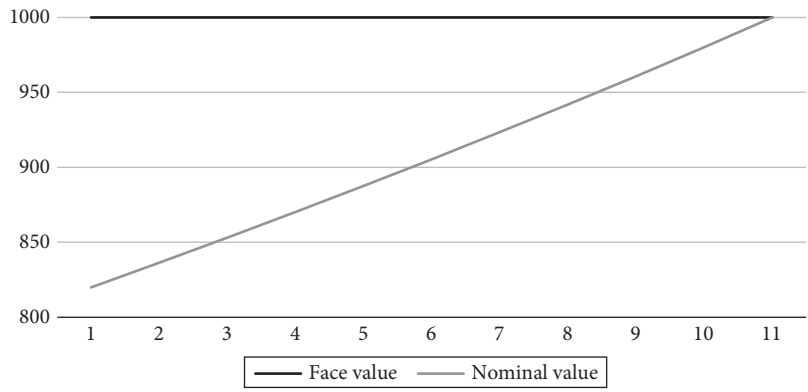

Figure 2.4 Face value vs. nominal value

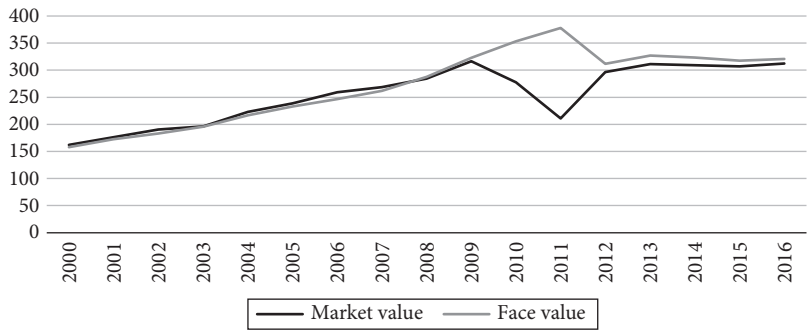

Figure 2.5 Greece central government debt securities, € billions
Source: IMF Government Finance Statistics.

issuance of different kinds of debt instruments that get recorded in different ways in official debt statistics, means users need to be aware of the basis of valuation of debt, and the types of debt being issued.

Notwithstanding the differences between face and nominal value because of differing instruments, differences between face or nominal and market value can emerge, either during periods of market panic (as investors fret that a government may default) or conversely during a flight to safety, when the price of government assets are bid upwards by investors seeking safe haven investments.

Figure 2.5 shows the first scenario, in relation to Greece. During the global financial crisis, and at the height of the Greek debt crisis in 2010 and 2011, the market price of Greek government debt securities plummeted, even as debt at face value continued to rise.

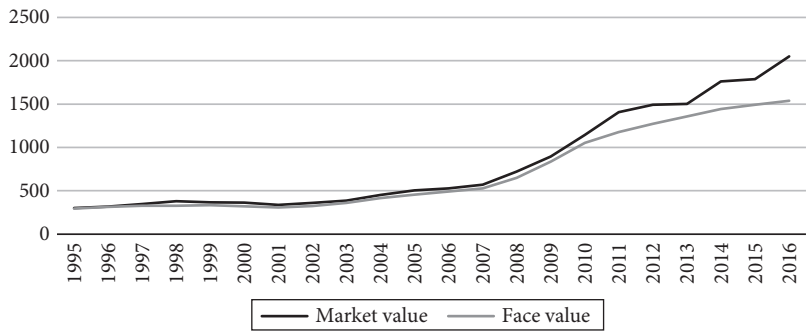

Figure 2.6 UK central government debt securities, £ billions
Source: IMF Government Finance Statistics.

Figure 2.6 shows the second scenario. Throughout the 1990s until the financial crisis, UK debt securities were little different at face and market value, but since the financial crisis a global shortage of safe assets, coupled with QE in the UK, has seen the market price for UK gilts increase, such that by the end of 2016 UK central government debt securities were worth considerably more at market value than at their face value. The same thing has happened in the United States.

Valuation is critically important to measures of sovereign debt, and users of debt data should aim to understand what valuation is being used, for which instruments and the extent to which a different valuation could provide different insights.

C. Instrument coverage of debt

Discussion of sovereign or government debt has typically focused on two main types of borrowing by governments: borrowings in the form of debt securities, such as US Treasuries, German Bunds, or UK Gilts, and borrowings in the form of loans, including loans from domestic and foreign commercial banks, and bilateral loans from foreign governments and their lending arms (such as development banks), as well as from international financial institutions such as the European Investment Bank, or World Bank.

Debt securities are the most common form of debt for most advanced and emerging market countries. Typically issued by public auction, a debt security is a promise to repay an amount at maturity, and will typically include a fixed interest rate, regular interest payments (coupons) and they come in many different varieties. Debt securities can be short-, medium-, or long-term

instruments (securities of under 1 year are typically referred to as Treasury Bills), can be issued at their face value ("at par"), for less than face value ("at a discount"), or for more than their face value ("at a premium").[9] Some instruments are index linked (either the coupon, the principal, or both) to the value of the Consumer Price Index, or other indices. Debt instruments can be issued in domestic or foreign currencies. Finally, debt securities can be issued to domestic creditors, or to external creditors, for example through the Eurobond market.

As we will discuss further below, the nature of bonds issued has an impact on the measurement of debt, depending on how debt is being valued. In addition, as discussed in Section 4, the maturity profile, currency of issuance, and residency of creditor can all have important implications.

While almost all countries can issue short-term securities in domestic markets, for many low-income countries, especially those with poorly developed or illiquid financial markets, or indeed no access to credit markets, debt securities are not a sufficient financing option. Instead for many low-income countries most of their debt is in the form of bilateral loans. These can be from domestic or foreign banks, or other financial institutions, but can also be highly concessional loans from bilateral or multilateral lenders. This difference between those more advanced or developed countries that mostly finance themselves with debt securities, versus the smaller or less developed countries that are more dependent on loan financing is shown in Figure 2.7.

The fundamental distinction between a debt security and a loan is that debt securities are designed to be tradable or *negotiable* whereas loans typically are not, so debt securities will often be redeemed by someone other than their original purchasers.

Government's (and corporations) are also continually coming up with new schemes to deliver public policy that can have the effect of hiding debt off the balance sheet, and whether intentional or not, these schemes can massage the publicly available debt numbers. To take one well-known example, during the late 1980s, governments in the UK and United States began to develop a new model for the construction of public infrastructure, the public private partnership (PPP). Under this model a private sector partner builds an asset and leases the asset to the government over a long-term contract. The government gets the benefit of a new hospital, prison, or school, but the debt is legally incurred by the private partner and no debt appears on the government

[9] Face value refers to the stated value of a bond stated by an issuer, for bonds this is the amount paid at maturity.

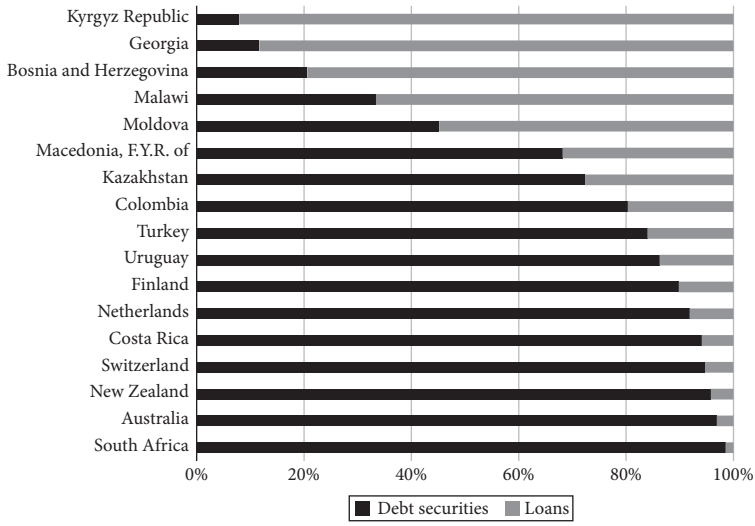

Figure 2.7 —Share of debt securities vs. loans selected countries—central government 2016
Source: IMF Government Finance Statistics

balance sheet. Today there are thousands of PPP style contracts across dozens of countries, which have funded billions of new infrastructure. New guidance has been developed, by the IMF, by Eurostat, by the International Public Sector Accounting Standards Board, and national authorities to provide guidance on whether to record these contracts as government liabilities, essentially as finance leases, and therefore a form of loan, but many countries do not follow these rules in their own debt statistics.

Some measures of debt also go beyond debt securities and loans. Maastricht Debt, the headline measure of government debt in the EU discussed earlier, includes not only debt securities and loans, but also government liabilities in the form of currency and deposits. The size of these liabilities varies across the EU, with currency and deposit liabilities of less than 0.5 percent of GDP in Belgium, Croatia, Hungary, Netherlands, Poland, Slovenia, Slovakia, and Finland, to countries which have significant general government currency and deposit liabilities including Ireland (2017—8.4 percent of GDP), UK (9.7 percent), Italy (14 percent), and Portugal (16.3 percent). Where measures of sovereign debt extend beyond the general government to include the wider public sector, including public sector banks or the Central Bank, the inclusion or not of currency and deposit liabilities can have a material impact on headline debt numbers (see Case Study 2.2).

Case Study 2.2 Currency in circulation and Central Bank reserve liabilities—debt or non-debt?

A significant form of balance sheet liability in most countries is the Monetary Base, comprised of currency in circulation (notes and coins) and central bank reserve liabilities, but is this really debt?

To the extent that debt is only compiled for general government, and excludes the Central Bank, this question can be sidestepped. But if you are compiling public sector debt, as is the case in several countries, then this is a valid question.

The treatment of notes and coins as a liability and as a debt of the issuer is arguably a legacy of the gold standard, when notes and coins were convertible to gold. But in today's fiat currency world, this is obviously no longer the case, and there is a good argument for treating notes and coins like gold—as a financial asset with no matching liability.

Whether currency in circulation is debt matters, clearly, but it would be less of an issue if all countries had similar amounts of currency in circulation. In reality it varies considerably in size. At one extreme are cash reliant countries with poorly developed banking systems, such as Afghanistan, Algeria, or the Kyrgyz Republic, with currency in circulation of 16, 24, or 19 percent of GDP respectively. At the other extreme are advanced economies with highly developed cashless economies, such as Denmark or Sweden, where the currency in circulation is just 3.3 or 1.3 percent of GDP respectively. But there are exceptions to this general rule, such as Japan. Japan has currency in circulation of over 20 percent of GDP. This is not due to a lack of development of Japan's banking system, but cultural factors including low crime rates, high levels of trust, and a long-standing preference for cash (Figure 2.8).

Central Bank reserve liabilities, the other major component of the Monetary Base, raise similar questions about whether they are really debt, especially in those countries with actual or de facto reserve requirements. While these appear on the Central Bank balance sheet as currency and deposit liabilities, and therefore would appear in a broadly defined measure of public sector debt, they are not the same as debt securities or loans, and again, vary significantly in size. Whereas the average central bank reserve liabilities are around 10 per cent of GDP, in some countries they are considerably higher, at 46 percent of GDP in the Czech Republic, and 67 percent in Japan (Figure 2.9).

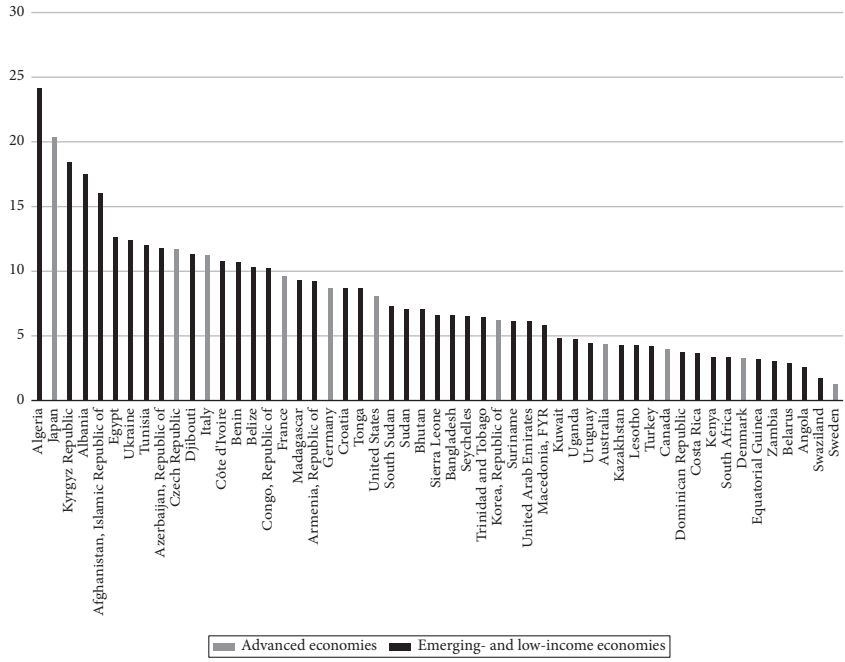

Figure 2.8 Currency in Circulation end-2017 % of GDP

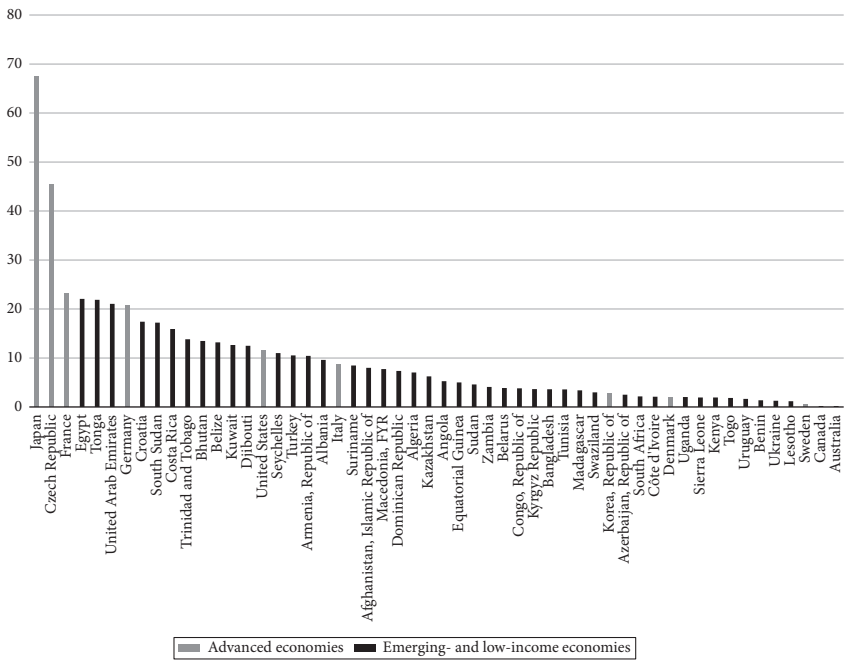

Figure 2.9 Central Bank Reserve Liabilities 2017 % of GDP

Continued

Case Study 2.2 *Continued*

Finally, the question of whether central bank reserve liabilities are debt or not raises questions about the impact on debt of Quantitative Easing (QE). Over the last decade, since the financial crisis, Central Banks in Japan, the UK, and the United States have acquired significant holdings of their own government bonds. While this has not reduced government debt, the composition of public sector debt has changed, reducing the amount of debt in the form of debt securities to a much higher proportion of central bank reserve liabilities. As a result, if you were to use a broad sectoral coverage of debt that included both the government and the central bank, but a narrow instrument coverage of debt (just debt securities and loans), then QE would have resulted in a lowering of public sector debt. Using a narrower sectoral coverage, or broader instrument coverage would mean QE had no impact. Again, this demonstrates how much these factors matter to our understanding of sovereign debt.

Although the traditional focus for sovereign debt is on debt securities, loans, and in some cases currency and deposits, the accounting and statistical communities recognize that a government can incur additional liabilities, such that focusing just on debt securities and loans could mean missing considerable government liabilities (Figure 2.10).

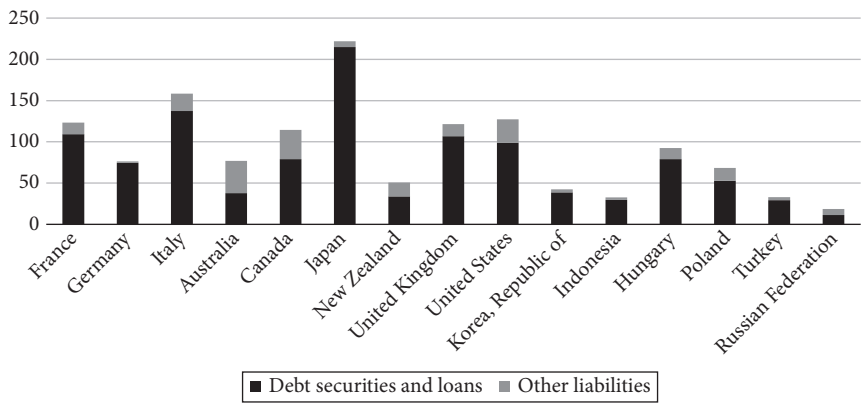

Figure 2.10 Debt Securities and Loans vs Other Liabilities (2016) % of GDP

Examples of other liabilities, for government and the wider public sector include accounts payable, financial derivatives, and, in many countries, pensions. These wider liabilities are captured in more comprehensive measures of the government or public sector balance sheet, discussed in Section 3.

3. Beyond debt: non-debt, contingent liabilities and public sector assets

Sovereign liabilities extend beyond the standard measures of sovereign debt discussed in the Section 2, further complicating the picture. This section aims to shed light on the main liabilities beyond debt. Specifically, it will discuss non-debt liabilities and contingent liabilities and liabilities in the SOE sector, outside the perimeter of general government. To capture all these liabilities, independent on whether they are recorded inside or outside of general government, we take the entire public sector as the point of departure.

This section further broadens the scope of analysis to include public sector assets, focusing on the public sector balance sheet. Discussions of sovereign liabilities focus on an important part of the government balance sheet, but this is only part of the story of a government's financial health. Financial statements for a multinational enterprise or other large company will typically include not just information on debt, but also the company's assets. In fact, its entire balance sheet, as well as an income statement, cash flow statement, statement of changes in equity, and extensive notes to the accounts with disclosures on material risks, including contingent liabilities, are normally published. Similarly, the Public Sector Balance Sheet (PSBS) brings together all the accumulated assets and liabilities that the government controls. It extends the perimeter of coverage from the general government to the entire public sector, bringing in nonfinancial and financial public corporations, including the central bank. As such, it presents the broadest possible picture of the health of public finances. Compiling this data through time provides clarity on trends and aids analysis and understanding.

A. Non-debt liabilities

Besides traditional debt liabilities (debt securities and loans), government or public sector liabilities also include other "non-debt" liabilities. These consist mainly of other accounts payable, financial derivatives, or, most significantly,

pensions. In short, narrowly focused debt statistics that only include traditional debt securities and loans may not tell the full story of government debt. Considering these liabilities other than debt securities and loans reveals a more comprehensive coverage of what governments or the public sector ultimately owe.

Currency and deposits are an integral part of the PSBS. These include the deposits held in public sector banks, reserves held with the Central Bank, and currency issued by the bank. In some cases, government itself has currency and deposit liabilities, for example when state-owned enterprises deposit their own funds in the government's single treasury account. The European Union's Maastricht Debt includes liabilities in the form of debt securities, loans *and currency and deposits*, and while currency and deposits are small in most EU member states, in 2016 they were much more significant in Ireland (8 percent of GDP), Italy (14 percent), Portugal (15 percent), and the UK (9 percent).

The public sector also carries amounts of accounts payable, short-term obligations to pay suppliers or other creditors. When expenditure is recorded on an accrual basis, any expenditures recorded where payments have not yet been made are recorded as giving rise to an account payable on the government balance sheet. These vary significantly in size, at the general government level from 3 percent of GDP in Latvia, to double digit amounts in France (12 percent) or Canada (18 percent). For countries that record their government spending on a cash basis, not recording accounts payables in the stock of debt enables them to run up, but not reveal, mounting domestic arrears on these payables, that would be recorded as expenditures and liabilities in accounts payable under accrual based accounting, that most governments require their private sectors to produce.

Pension liabilities can be as large or larger than traditional concepts of sovereign debt. And pension obligations are often—at least in part—enforceable, making these obligations very similar to debt, even though they are not considered as such in most countries. Public sector pension obligations can be related to unfunded pensions schemes managed by the government or to shortfalls for funded schemes for public sector employees. These liabilities, which may be difficult to measure, can be very large. At the end of 2015, Belgium estimates that employment related pension liabilities for only government employees stood at €181 billion, or 44 percent of GDP.

In addition, under most international statistical and accounting rules, the public sector balance sheet does not even include liabilities of social security, state, or national pensions that apply to the whole population and which are funded on a pay-as-you-go basis or out of general taxation. This is because

such social security schemes are not deemed a contractual obligation by statistical and accounting guidance and can often be changed or amended by the government. However, for many households there are strong expectations that these will be paid, especially those households already in retirement. In 2015 Germany estimated that liabilities of its social security schemes totaled some €6.8 trillion, or around 226 percent of GDP. Figure 2.11 shows the value of accrued to date liabilities of unfunded, employment-related pensions and social security pensions in eight EU member states, at the end of 2015. These types of figures, if included inside gross debt, would dramatically change perceptions of gross debt for these countries, anchored as they are by the Maastricht Debt threshold of 60 percent of GDP.

Valuation matters with pension liabilities as much, if not more, than with debt liabilities. Pension liabilities, whether employment-related pensions on the government balance sheet or social security obligations disclosed elsewhere, are difficult to value. They are conventionally valued by actuaries as the discounted value of future pension payments. This means bringing together real information on the participants in a pension scheme, with assumptions about future mortality rates, wage growth, and a discount rate. Small changes in these assumptions, especially the discount rate, can have significant impact on the resulting liabilities.[10]

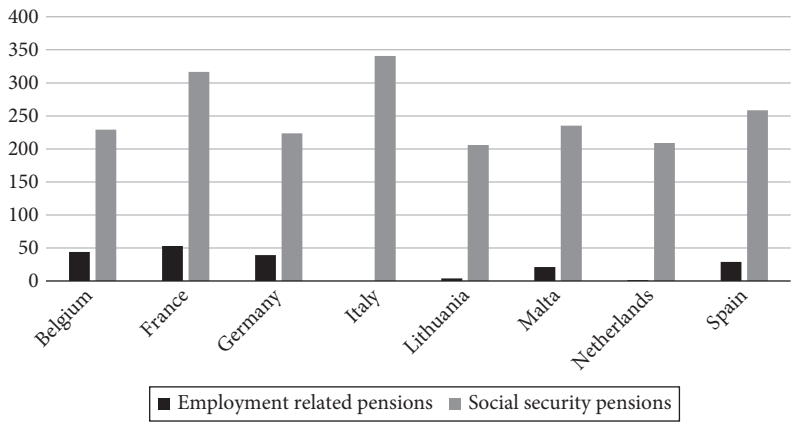

Figure 2.11 Accrued to date liabilities—governmment unfunded employment-related and social security pensions (2015) % of GDP

Source: Eurostat.

[10] EU countries are required to compile estimates for accrued to date liabilities of their various national pension systems, using a unified and consistent set of assumptions including a discount rate of 5 per cent. However, countries are also asked to conduct sensitivity analysis, calculating the

Leaving pensions off the balance sheet, or treating them as contingent liabilities, does not prevent pension chickens coming home to roost. In Brazil, pensions are widely considered to be an explosive and possibly unsustainable government liability. General government expenditure on social benefits, including pensions, have gone from 10.2 percent of GDP in 2006 to 17.6 percent of GDP in 2017 and are projected to continue climbing absent reforms of the pension system. The traditional focus on debt and the primary balance revolves around concerns of the amount of expenditure on interest and crowding out of other spending. In Brazil the costs of servicing this different kind of liability, the pension liability, is a major source of fiscal concern. Brazil faces a further problem when dealing with this issue, which is that pensions can only be modified with a constitutional amendment.

Other countries have recognized the issues posed by their pension systems, and engaged in pension reform. In some cases, when countries reform their pension systems, the reforms can turn the unfunded obligations of a pension system into something that looks a lot like traditional government debt liabilities. The best example of this is from Chile in the early 1980s. Faced with an increasingly underfunded social security system, the Chilean government moved to a system of individual pension savings accounts, but to reflect rights built up under the old system, retirees were able to claim so called "recognition bonds" that make explicit the debt owed to workers who had contributed to the old pension system. While not included in headline debt figures, information on recognition bonds is included in Chile's Reports on Public Debt Statistics.[11] Chile's stock of recognition bonds was 37.9 percent of GDP in 1982. Recognition bonds fell to below 10 percent of GDP in 2006 and are now just 0.8 percent of GDP at September 2018.

Finally, some governments also manage their debt using a final type of instrument, a financial derivative. While not common, and not a large part of debt, some government debt management offices use hedging instruments to manage exchange rate or interest rate risk. But while the use of derivatives may be limited, depending on the type of derivatives issued, they may contain considerable fiscal risk.

liabilities using a 4 and 6 percent discount rate. The impact is significant. In Ireland, as an example, using a 5 percent discount rate, government employment related pensions and social security schemes liabilities at the end of 2015 stood at €345 bn (132 per cent of GDP). Using a 6 percent discount rate, liabilities fall to €284 bn (108 per cent of GDP). By contrast using a 4 per cent discount rate would see liabilities rise to €424 billion, (162 per cent of GDP).

[11] http://www.hacienda.cl/english/public-debt-office/statistics/public-debt.html

B. Contingent liabilities

Some sovereign debt definitions also include contingent liabilities, such as publicly guaranteed debt. Contingent liabilities are a possible obligation depending on whether some uncertain future event occurs.

The balance sheet approach can also help to bring out contingent liabilities. Extending coverage to the public sector brings contingent liabilities from public corporations into the balance sheet. However, a range of other contingent liabilities remain outside. These include explicit government guarantees or other contingent risks from the private financial sector (such as guarantees of the deposit protection schemes or implicit guarantees for "too big to fail institutions"), PPPs, natural disasters, and legal risks. The potential impact from these contingent liabilities on public finances can be informed by other assessments, such as FSAPs, fiscal transparency evaluations, and countries' fiscal risk statements. Bringing out these risks of the realization of contingent liabilities on public finances can provide guidance on the size of buffers that may be needed to avoid pro-cyclical policy adjustments during a crisis, the channels through which fiscal risks propagate, and where risk management efforts should be directed (IMF 2016a).

Contingent liabilities can be large and consequential but may be hard to quantify precisely ex ante (see Case Study 2.3). In some cases, the government's

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Case Study 2.3 How contingent liabilities gave the Celtic Tiger a heart attack

At the start of the 1990s, Ireland was a poor country by West European standards, with high poverty, unemployment, inflation, and low growth.[1] Then, something dramatic happened starting in the mid-1990s. Between 1995 and 2000 the Irish economy expanded at an average rate of 9.4 percent and continued to grow at an average rate of 6 percent until 2007 earning the country the nickname the "Celtic Tiger." In 2008, the Tiger had a massive heart attack.

Starting in 2008, Ireland had to contend with an interlocking sovereign–banking–real economy crisis. After facing heavy losses on property-related

[1] This description of the Irish case is based on IMF reports. It borrows heavily from the 2012 Article IV report (IMF 2012). Other key sources included Ireland's Fiscal Transparency Assessment (IMF 2013) and IEO (2016).

Continued

Case Study 2.3 *Continued*

assets in the spring of 2008, Irish banks suffered a run on wholesale funding in the Fall—prompting massive recourse to Eurosystem liquidity support.

In response, the government issued a blanket guarantee from September 2008, transferred large distressed property development and commercial real estate assets from banks to the National Asset Management Agency (NAMA) from April 2009, and provided large scale support for two failed banks (Anglo Irish and Irish Nationwide Building Society), and large equity injections in other banks (IMF 2012).

The materialization of these contingent liabilities together with a full-fledged economic bust ultimately led the authorities to request a bailout. A steep decline in construction activity drove the country into recession from 2008 with the sharp world trade contraction in 2009 adding to the shock to Ireland's highly open economy. The fiscal deficits ballooned, and public debt shot up from 25 percent of GDP in 2007 to over 90 percent by 2010 (IMF 2012). Confidence in the country's fiscal position crumbled, primarily because of the close sovereign–bank interlinkages. Deepening uncertainty about the ultimate scale of the banking sector losses, and hence growing doubts about public debt sustainability, drove a brutal switch in market sentiment in the Fall of 2010, cutting the sovereign off from market financing. In December 2010, the Irish government requested EU and IMF financial support. The total financing package of euro 85 billion (about US$113 billion at the time) was provided jointly by the EU Financial Stabilization Mechanism/European Financial Stability Facility, bilateral partners, the IMF, and the government's own resources. With an estimated fiscal cost of some 40 percent of GDP, Laeven and Valencia (2012) consider Ireland's banking crisis the second costliest in advanced economies since at least the Great Depression.

exposure to contingent liabilities can be quantified.[12] In others, quantification may be hard (open-ended schemes) or best not be published, in order not to affect the government's negotiation position. The IMF Fiscal Transparency

[12] See, e.g., Igan et al. (2019) or Laeven and Valencia (2008, 2013, 2018) for ex-post quantifications of public interventions in the financial sector during and after the global financial crisis.

Code recommends governments analyze and disclose potential risks through Fiscal Risk Statements (see Annex 2.B) or similar publications.

In addition to explicit contingent liabilities, there are often implicit liabilities, such as political or public pressure to stand behind certain institutions, even for liabilities that are not explicitly guaranteed. In the United States, although there were many explicit statements to the effect that the debt of Fannie Mae and Freddie Mac were not guaranteed by the US Federal Government, prior to the crisis these two mammoth government sponsored entities were issuing AAA rated mortgage backed debt securities (backed by increasingly risky underlying loans), despite being considerably riskier propositions than the US Government and at the height of the financial crisis these two units, both far too big to fail, were ultimately rescued by being placed into "conservatorship" by the US Treasury.

C. Public assets, net debt, and net worth

In 2016, Norway's general government debt was 36 percent of GDP, low compared to many other advanced economies. However, Norway's net debt was –85 percent of GDP, meaning Norway didn't have net debt, but instead it had net assets of 85 percent of GDP. In fact, Norway's financial situation is rosier still, Norway's net financial worth (all financial assets minus all liabilities) is 290 percent, and Norway's net worth (including all nonfinancial assets) is over 350 percent of GDP. While Norway is an outlier, the traditional focus on government liabilities is nowhere near the whole story of a government's financial position.

An assessment of public wealth and the health of public finances should look beyond debt. To be comprehensive, it should incorporate government assets and non-debt liabilities, as well as assets and liabilities of the broader public sector outside of general government. This section will focus on these elements of the public sector balance sheet.

Economist Paul Krugman often compares the US Federal Government to an insurance corporation with an army; leaving aside the military, any analysis of an insurance corporation needs to understand not just the liabilities it has incurred, but also the assets it has amassed to meet any claims. Consequently, to fully understand a government's financial position one should look at *its* entire balance sheet, much like one would look at a corporation's balance sheet to assess its financial position. Looking beyond sovereign debt at a country's PSBS brings out government assets and non-debt liabilities, as well as the

assets and liabilities of public corporations.[13] This strengthens fiscal analysis, showing the full scale of assets and liabilities that the government controls. Moreover, in many countries public corporations represent a significant source of fiscal risk, either through explicit or implicit guarantees, or, in some cases, a direct draw on the public purse (Bova et al. 2016).

These items are material, with public sector assets comprising US$101 trillion or 219 percent of GDP in a sample of thirty-one countries covering over 60 percent of world GDP (Figure 2.12). The PSBS also provides a complete picture of liabilities, illustrating that general government debt comprises 94 percent of GDP. But that is only half of the total public sector liabilities of 198 percent of GDP. Assets include financial assets and nonfinancial assets. Public corporations include both financial and nonfinancial corporations, with the Central Bank included in the former.

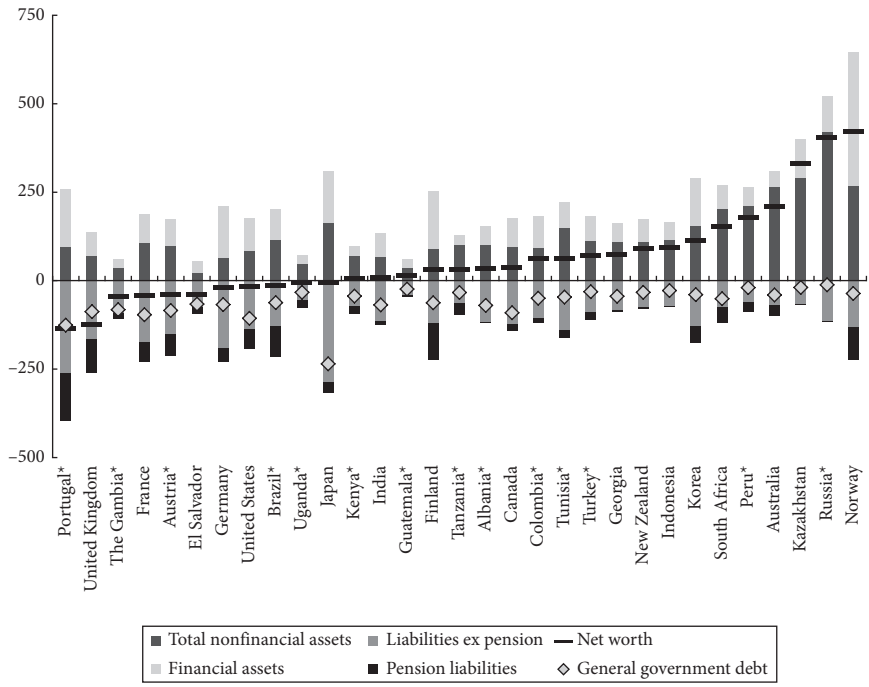

Figure 2.12 Public sector balance sheets, 2016 (% of GDP)

Source: IMF staff estimates.

*Based on a single year of data, in most cases compiled as part of the Fiscal Transparency Evaluation: Albania, 2013; Austria, 2015; Brazil, 2014; Colombia, 2016; The Gambia, 2016; Guatemala, 2014; Kenya, 2013; Peru, 2013; Portugal, 2012; Tanzania, 2014; Tunisia, 2013; Turkey, 2013; Uganda, 2015.

[13] The equity value of public corporations is included within general government accounts, as part of financial assets. Thus, the inclusion of public corporations within the public sector has no impact on net worth.

Elements of the PSBS

Financial assets typically include cash deposits, government loans to other sectors, as well as equity in public and private corporations, as well as debt security holdings of pension and sovereign wealth funds. These assets may be marketable and relatively liquid (particularly if they are listed and traded in deep markets), but they may also be hard to value (such as equity in state-owned enterprises) or hard for the government to monetize (if the assets are explicitly tied to pension, social security, or other obligations, or held by subnational government units they may not be available to finance other funding needs). Some financial assets can be relatively volatile, due to substantial revaluations as asset prices fluctuate.

Non-financial assets typically include buildings, infrastructure, land, and natural resources. Many of these comprise the public capital stock and play an integral role in delivering economic and social outcomes. Existing government data are often missing or poorly reported, with serious valuation issues (Bova et al. 2013). For commodity producers, natural resources can represent the largest asset on the state's balance sheet (see Figure 2.13).

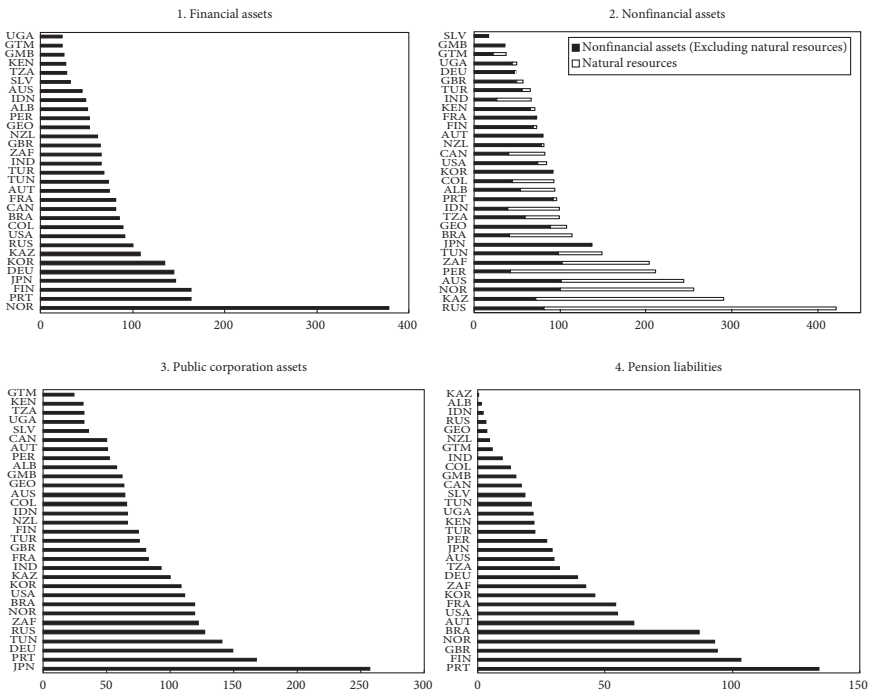

Figure 2.13 Additional elements of the public sector balance sheet (% of GDP)

Source: IMF 2018 as in original chart

Including natural resource assets introduces a greater rigor to the management of the public's wealth. In standard fiscal analysis, sales from natural resource extraction are treated as a revenue, increasing the net worth position of the state. The balance sheet approach instead recognizes them as an asset, which once extracted and sold represents a conversion of one asset (resources) for another (cash). Apart from extraction costs, this conversion is net worth-neutral. The ultimate impact on public wealth is then determined by what the government does with the cash receipts. If revenues from sales of natural resources are used to fund ongoing expenditure, net worth decreases, whereas if they are used to purchase other assets (financial or non-financial), net worth remains broadly unchanged. Estimates used here for the stock of mineral and energy resources correspond to the net present value of the expected pre-tax cash flows resulting from their commercial exploitation.

Consolidation

Extending the perimeter of the balance sheet to the public sector requires a consolidation of cross holdings of assets and liabilities by different public sector entities.[14] These are country specific, but the largest cross holdings are typically government deposits at the central bank, financial corporations' holdings of government securities, the government's equity stake in public corporations, and loans between public corporations. Consolidations can be large, and potentially alter the fiscal picture. For example, in Japan, while gross outstanding public sector debt securities and loans were worth 288 percent of GDP in 2017, the majority of this is held by other public sector units, leaving 138 percent of GDP in the hands of private creditors (Figure 2.14). The same is true in the United States, to a lesser extent, where the equivalent figures are 164 and 110 percent of GDP.[15] These cross holdings can be a channel through which fiscal risks spread. For instance, during downturns or crises, public corporations often build up large arrears to each other, leaving a trail of unpaid bills across the sector and clogging up balance sheets. Japan presents an interesting example of intra-public sector links that triggered a policy response. Until the year 2000, the Postal bank was required to lend its deposits to the Fiscal Loan Fund (Figure 2.14, bottom). Partly as a result, the Postal bank experienced losses when interest rates declined in the mid-1990s, due to a mismatch

[14] Consolidation refers to the removal of intra-public sector claims. Consolidated public sector liabilities thus represents the total amount of liabilities the public sector is standing behind, when all cross holdings have been netted out.

[15] In the United States, treasury holdings by the social security fund are consolidated within the general government and hence do not show in these numbers.

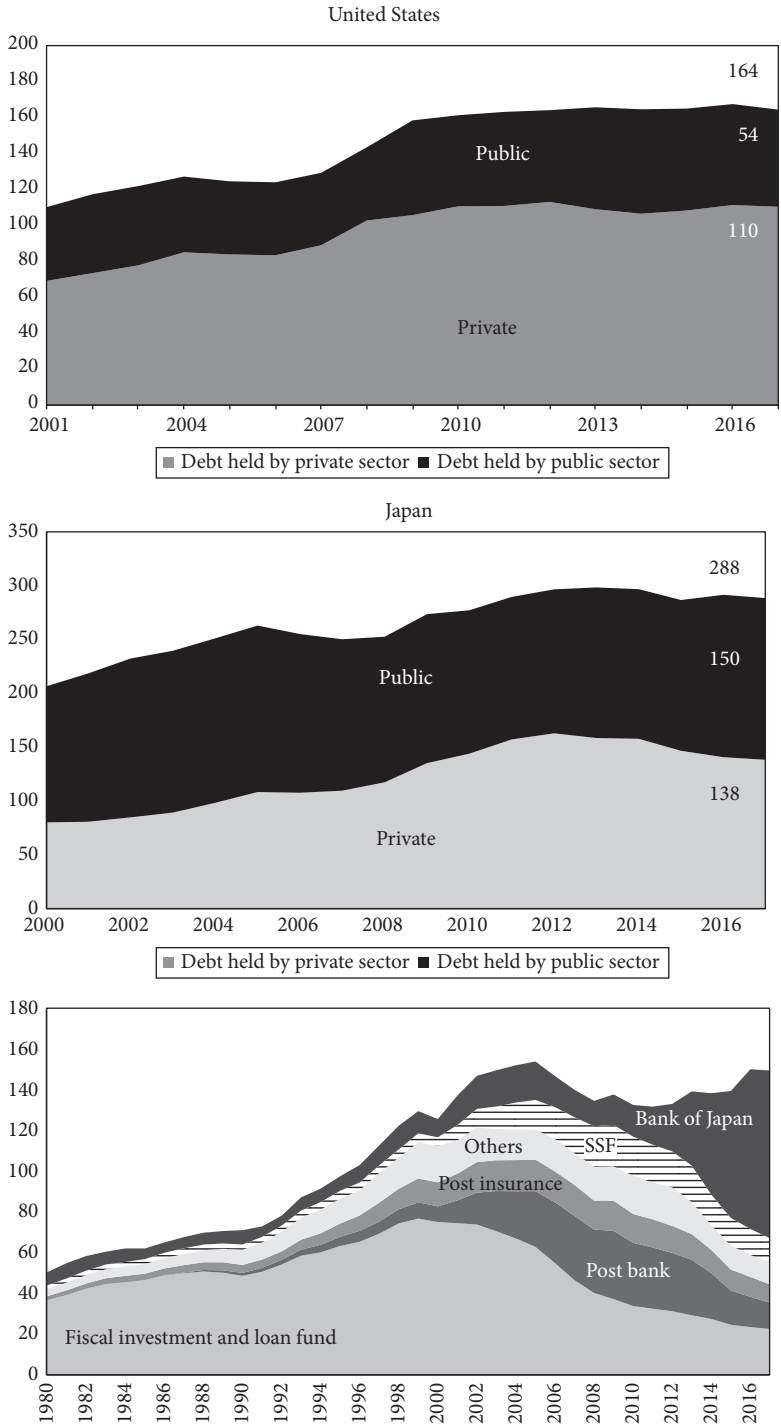

Figure 2.14 Government debt held by public and private sector (% of GDP)

between its long-dated liabilities and shorter-dated assets. Meanwhile, the Fiscal Loan Fund realized offsetting profits. While on net these profits and losses hence cancelled out across the public sector, concerns over Postal bank's losses triggered a reform of the system (Koshima 2019).

Valuation

As with the valuation of debt and pensions, the valuation of assets matters, and is not straightforward. Valuation can be a challenge, particularly for non-financial assets that are rarely traded, and with differing approaches taken for different components of the balance sheet across countries—in part due to differences in accounting standards. Asset valuations are also more volatile than debt and can be highly correlated with the economic cycle—meaning their values can be at their nadir when financing needs are most pressing.

Liquidity

In addition, many assets are illiquid or not marketable, and would not be available to meet rollover or deficit financing needs in the short term. Financial assets are mostly marketable and relatively liquid, except for direct loans and non-listed equity holdings in public corporations, which may also be less reliably valued. However, some financial assets may be explicitly tied to pension or social security obligations and may not be available to finance other funding needs. Nonfinancial assets include buildings, infrastructure, and land. They are often illiquid and non-marketable, or only marketable over the medium to long-run (e.g., privatizations).

Public wealth

Taking assets on board creates a comprehensive view of public wealth. The main indicators used to bring this out are net debt, net worth, and net financial worth:[16]

- *Net worth*, the headline measure of government wealth. It is calculated as total assets minus total liabilities. Net worth suffers from the various valuation issues that accompany the constituent parts of the balance sheet, particularly from nonfinancial assets. Furthermore, it makes no distinction between assets that can be sold to meet financing needs, and assets that are not marketable.

[16] Besides net (financial) worth, a range of other indicators provide important information on the state and resilience of public wealth. These include the standard measure of gross debt, as well as measures that explore mismatch risks and degree of hedging present in the balance sheet.

- *Net financial worth*, a measure of financial wealth, calculated as total financial assets less liabilities. In general, financial assets and liabilities are more reliably valued, and more readily marketable than nonfinancial assets.
- *Net debt*, a narrow measure of wealth, calculated as debt liabilities (most commonly debt securities, loans, and currency and deposits), minus corresponding financial assets. This definition leaves out hard to value assets, such as government equity holdings in state-owned enterprises.

The evolution of net worth presents a somewhat different picture from the more standard evolution of debt. A decomposition of post-crisis developments in a selected sample of countries shows the relative importance of debt accumulation, public investment, operations in the public corporation sector, and valuation changes (Figure 2.15). It shows that while deficits and debt accumulation drove the post-crisis deterioration in solvency, balance sheet effects significantly cushioned the decline in net worth. Specifically, among these countries, net worth has declined by some 25 percentage points of GDP since the crisis. Fiscal deficits are by far the largest component of this decline, contributing 38 percentage points of GDP to the overall decline. Together with the 9 percent of GDP denominator effect this bring net worth

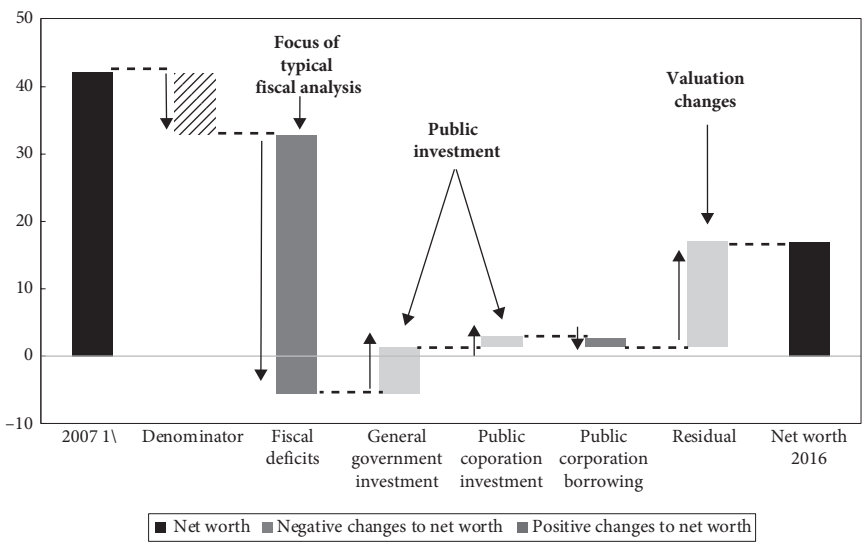

Figure 2.15 A decomposition of changes in net worth, weighted average of 17 countries, (% of GDP). 1\Expressed as % of 2007 GDP.
Source: **IMF (2018)**

down into negative territory.[17] However, some of the deficits were used to invest rather than consume, raising net worth by some 8 percentage points of GDP. While valuation fell during the crisis, reflecting falling asset prices, they rebounded in subsequent years, adding another 16 percentage points of GDP to net worth. Hence, overall the decline in net worth has been far smaller than the cumulative deficits alone would suggest.

Overall, the public sector balance sheet complements the picture presented by government debt. It provides the most comprehensive view of public wealth. By broadening the focus, it sheds light on the assets governments control, as well as liabilities that receive scant attention in standard analysis. It thus complements data and analysis based purely on debt, and this way can enrich fiscal analysis as well as the policy debate.

4. Sovereign debt composition

A. Supply (issuance of sovereign debt)

Analysis of government or sovereign debt does not only focus on the total stock of gross or net debt. The characteristics of debt also matter enormously to whether a country has a debt problem or not. Analysis, therefore, will often look at the type of debt instruments (bonds versus loans), currency of issuance, the maturity profile, and the jurisdiction of debt issuance. Accordingly, this section discusses the typical structure of debt for advanced economies, emerging markets, and developing economies from a cross-country perspective. The discussion highlights important differences across income groups and countries in the composition of the debt they have issued.

Debt by instrument

As discussed in earlier, government debt typically consists of two main types of borrowing: borrowings in the form of debt securities and loans. While debt securities are the most common form of debt for advanced and some emerging market countries, loans are still the main form of borrowing for most developing economies (Figure 2.16).

[17] This denominator effect displays the impact of moving from 2007 to 2016 GDP in the denominator. The 2007 bar is expressed in percent of 2007 GDP, while all other bars are expressed in percent of 2016 GDP.

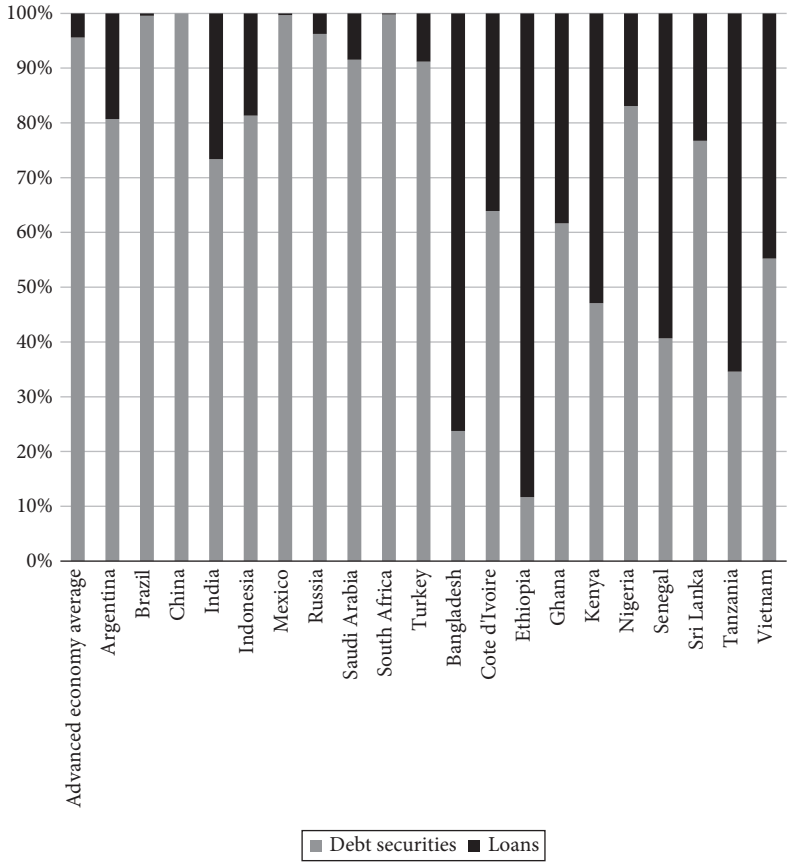

Figure 2.16 Composition of central government debt by instrument, end-2017

Note: Data for China and Nigeria are as of 2017Q2 and 2017Q1, respectively. Data for Saudi Arabia is for budgetary central government only

Sources: IMF/World Bank Quarterly Public Sector Debt (QPSD) database; national sources.

Debt by currency

Currency of issuance is another important dimension in debt structure. While large advanced economies almost exclusively issue debt in their own currencies, emerging markets, and to a larger extent developing economies, also borrow in foreign currencies (Figure 2.17). The borrowing is typically in US dollars, but are also in Japanese yen, euro, UK sterling or Swiss francs. While Argentina, Indonesia, Mexico, Saudi Arabia, and Turkey still have substantial government debts in foreign currencies (Figure 2.17), most emerging market economies have traditionally moved away from foreign to local currencies in recent years. By borrowing in foreign currencies countries may be

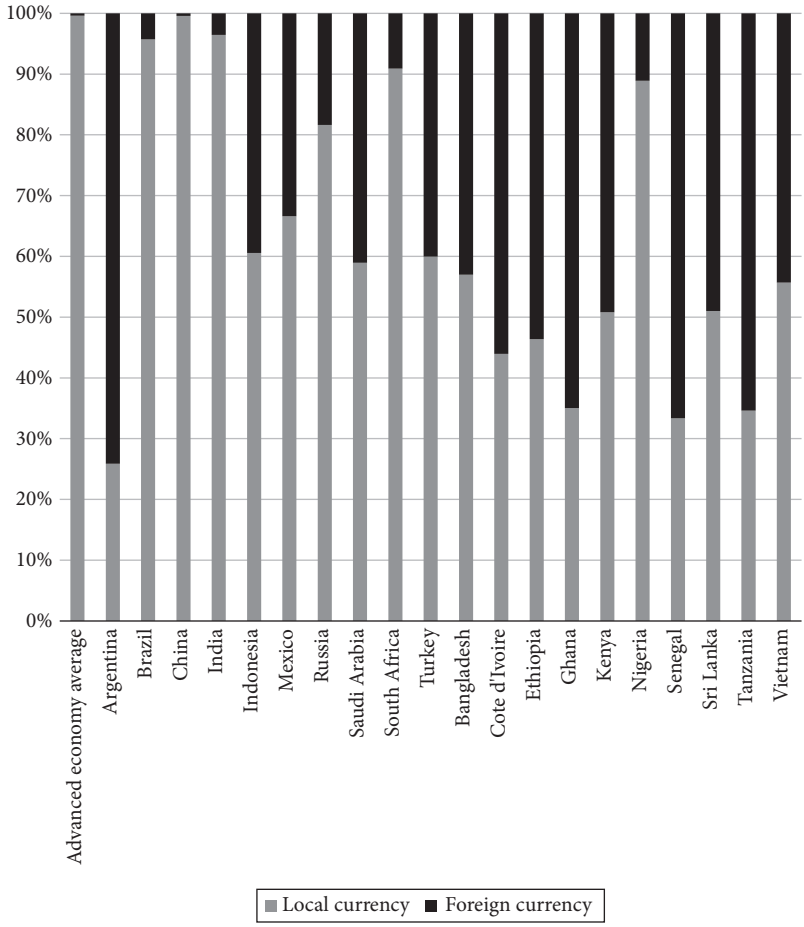

Figure 2.17 Composition of government debt by currency, end-2017

Note: Data for Nigeria are as of end-2016.

Sources: IMF DSA databases.

able to access deeper capital markets or borrow at lower headline interest rates but this exposes the borrower to what can be significant exchange rate risks.

Debt by maturity

Maturity profile is another important consideration. As discussed earlier, countries issue a mixture of short- and long-term debt, but the longer the maturity, the fewer short-term refinancing or rollover risks exist. As Figure 2.18 shows, today, most countries issue debt at longer-term maturities (i.e., greater than one year) to avoid short-term rollover risks. The share of short-term debt is typically no more than 10–20 percent for total government debt

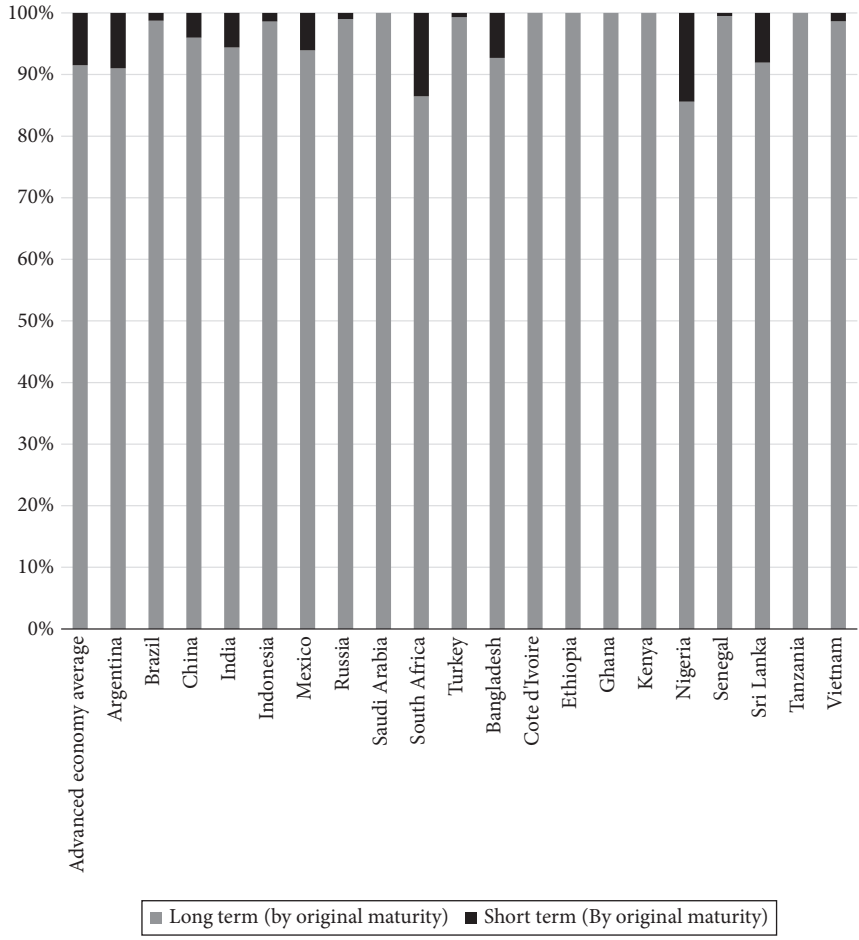

Figure 2.18 Composition of government debt by maturity, end-2017

Note: Data for Nigeria are as of end-2016.

Sources: IMF DSA databases.

outstanding (Figure 2.18). Countries where central banks issue their own debt securities do not need to rely on Treasury Bills for open market operations, but most central banks make use of Treasury Bills to manage liquidity and this creates a structural reason for more short-term government debt (Nyawata 2012).

Debt by jurisdiction of issuance

The jurisdiction of the debt has also major implications. Most advanced economies issue debt securities under their own domestic law but many emerging market and developing economies take advantage of international capital

markets and issue in the UK or United States under English or New York law. Examples include Argentina, Indonesia, Saudi Arabia, and Turkey among emerging markets and Cote d'Ivoire, Ghana, and Senegal among developing markets (Figure 2.19). While this typically involves a lower interest rate and access to a larger investor base, the sovereign submits to a foreign law and jurisdiction which may expose the sovereign to litigation risk. For the investors such submission to foreign law and jurisdiction results in larger protection and more predictability.

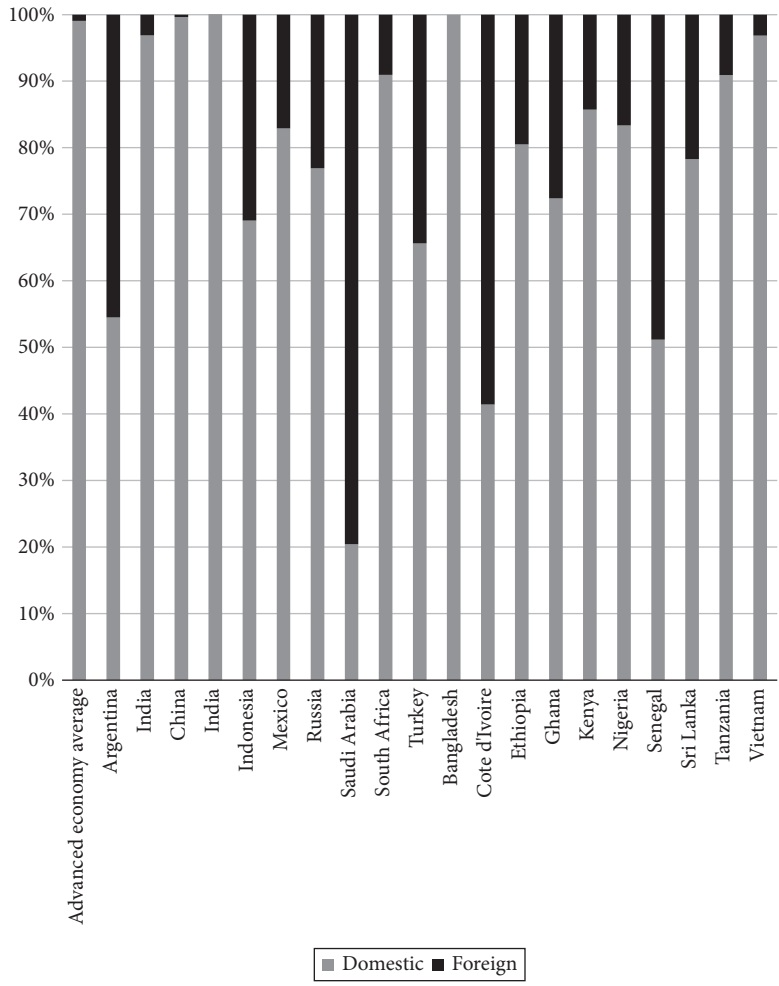

Figure 2.19 Composition of government debt by jurisdiction of issuance, end-2017

Note: Data refer to general government debt securities for advanced and emerging market economies; central government debt securities for developing economies.

Sources: BIS Debt Securities databases; national sources.

Finally, the residency of creditors can be important. This is discussed in more detail in Subsection 4.B which elaborates on the demand side/investor holdings of debt. Countries often issue debt both domestically and externally, but many analysts provide a much greater emphasis on the amount of external debt (see Case Study 2.4) and are much less concerned with the stock of

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Case Study 2.4 Different approaches to external debt

Reports and policy papers use the terms "domestic" and "external" public debt. What these terms mean can vary dependent on the context. Indeed, there are at least three concepts of defining whether a debt is domestic or external:

Residency. Following the balance of payment method, the first approach is to define external debt as a resident's liability to a non-resident. For instance, a German bank lending to say Brazil would be an external debt. A debt would be domestic if both parties, debtor and creditor, are residents of the same country (Gianviti 1989). Accordingly, a Brazilian subsidiary of a German company lending to Brazil would be a domestic debt.

Currency. A second approach is to distinguish the debt whether it is denominated in domestic or foreign currency. This is important since the sovereign needs to purchase foreign currency to service foreign currency debt. Foreign exchange debt also exposes the debtor to foreign exchange risks due to the devaluation risk of the domestic currency (Gianviti 1989). For countries using another country's currency as a domestic currency (for instance Kosovo using the euro or Ecuador using the US dollar) or for countries in a currency union (such as the euro area), the currency may be both a domestic and a foreign currency; however, the country in question has no or limited control over such currency.

Governing law. A third approach may be to use governing law as the deciding factor to determine whether the debt is external or international. A debt would be external if the liability is governed by the law of another country than the issuing country. This distinction is motivated by a recognition that, with respect to debt governed by domestic law, the legal leverage possessed by holdout creditors is more limited given the capacity of the sovereign debtor to modify its domestic law (IMF 2014). For instance, to facilitate the restructuring of its domestic law governed bonds, in 2012 Greece enacted legislation that aggregated claims across all the affected domestic law issuances, thereby eliminating the power of creditors to obtain a blocking position in an individual issuance.

Continued

Case Study 2.4 *Continued*

In practice, these concepts overlap. For instance, in most advanced countries, most of the debt is in domestic currency and issued under domestic law, but a good part is often held by non-residents (such as the United States, Germany). Emerging markets typically issue a mix of foreign and domestic currency debt, with domestic currency debt typically issued under domestic law, but foreign currency debt sometimes issued under domestic and sometimes under foreign law (such as Argentina). While foreign law debt is largely held by non-residents, the latter are increasingly participating in domestic law/domestic currency debt. For low-income countries, there is generally more alignment between the concepts: foreign currency debt is foreign law and non-resident held, while domestic currency debt is domestic law and resident-held (with exceptions for frontier markets such as Ghana, Figure 2.20). [reference to Chapter 9]

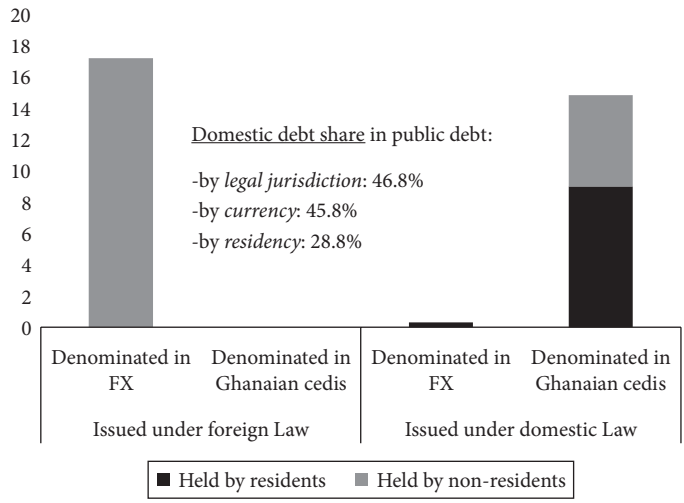

Figure 2.20 Breakdown of Ghana's public debt (US$ bn, end-2017)
Source: Ghana 2017 Annual Debt Management Report, and authors' estimates.

domestic debt (which may be held by domestic financial institutions and households for whom withdrawing capital is not as much of an option). Heavy reliance on external creditors exposes borrowers to risks related to the global capital markets (assuming the debt is commercial debt), and means creditors are exposed to shifts in investor sentiment, capital flight risks, and hot money flows all of which can come to a crunch during a crisis.

Over time, we have seen significant improvements in the composition of debt, especially for major emerging markets. Over the last decade or so, "dangerous" forms of debt (i.e., short-term and/or foreign currency debt) that increase the likelihood of sovereign debt crises, or render these crises more difficult to manage, have declined for many emerging markets. Similarly, issuance in domestic jurisdictions has increased for most emerging markets, allowing them to avoid some of the litigation risks associated with potential debt restructurings.

Specifically, as of end-2017, about two-thirds of emerging market debt securities are now issued in local currency (Arslanalp and Tsuda 2014b, updated). Similarly, as of end-2017, the average maturity of emerging market debt securities is now 8.3 years, up from 6.9 years a decade ago, according to Bank for International Settlements (BIS) figures. The BIS figures also show that emerging market debt securities are now predominantly issued domestically, with international debt issuance falling in relative terms (even if increasing in absolute terms).

Having said that, beyond these aggregate trends, there are still important differences in debt composition across emerging markets, and these have existed for some time. Guscina (2017) and Jeanne and Guscina (2006) document these differences for nineteen emerging markets during 1980–2012 and show that emerging Asia has had debt structures very similar to those in advanced countries, with a high share of long-term domestic-currency debt. In contrast, Latin America has historically had low shares of long-term domestic-currency debt. In particular, Guscina (2017) documents that domestic long-term local currency-denominated fixed-rate (DLTF) debt represents more than 80 percent of domestic debt for emerging Asia, as of end-2012. In contrast, the corresponding figures for emerging Europe Middle East and Africa (EMEA) and Latin America are about 60 percent and 30 percent, respectively. Those regional differences still exist to a large extent until today.

B. Demand (holders of government debt)

So far, we have discussed how to define and track the outstanding supply of sovereign debt over time. The demand side of government debt—who is holding the debt at any point in time—also matters greatly. In fact, events during the recent euro area debt crisis—and earlier during the emerging market crises of the 1990s—have illustrated that sovereigns, just like banks, can be subject to runs, highlighting the importance of the investor base of debt. Governments have seen sharp rises in borrowing costs or even lost

market access following dramatic shifts in their investor base, even when the supply of debt remained unchanged.[18]

While there are several ways to track the supply of debt, statistics on the investor base of debt are harder to come by. Public debt managers typically have only limited information on the ultimate holders of government debt securities once those get traded in secondary markets. For example, a debt manager may pay scheduled interest to an account maintained by Euroclear—a Belgium-based settlement and custodian company—but would not know whether the payment is for a commercial bank in Germany, a fund manager in Luxembourg, or a foreign central bank in Asia.

This data gap poses a risk factor. Fortunately, one can use a standardized approach to compile internationally comparable estimates of investor holdings of sovereign debt. The methodology, developed by IMF staff (Arslanalp and Tsuda 2014a, 2014b), facilitates tracking the investor base of more than US$50 trillion of sovereign debt on a quarterly basis starting from 2004. The estimates are constructed from publicly available international and national data sources and decompose the investor base into six types of investors—domestic central bank, domestic banks, domestic nonbanks,[19] foreign official sector, foreign banks, and foreign nonbanks. The breakdown allows for constructing risk indices—an investor base risk index (IRI) and a foreign investor position index (FIPI)—to assess a sovereign's vulnerability to a run by investors.

It is worth noting that domestic nonbanks, in many countries, include public sector units such as pension funds, provident funds, or sovereign wealth funds, that like the Central Bank may ultimately hold a lot of government debt. To take one example, in Singapore, at the end of 2017 the Central Singapore Provident Fund, a public unit classified outside of government as a domestic nonbank, holds special issues of Singapore government securities worth 77 percent of GDP. Total Singapore government gross debt was 108 percent, but the vast majority of this debt is essentially owed to one of its own public entities, and then to the citizens of Singapore in future pensions. This presents a very different scenario to other countries with high government

[18] In general, shifts in the sovereign investor base can (i) influence governments' borrowing costs; (ii) affect governments' refinancing risks; and (iii) create potentially harmful sovereign–bank linkages and threaten domestic financial stability, if domestic banks become highly exposed to own government debt.

[19] Nonbanks cover (i) institutional investors other than banks (i.e., insurance companies, pension funds, and investment funds) and (ii) households and nonfinancial corporations. While household or nonfinancial corporate holdings of government debt account for a sizable portion of nonbanks in some countries (Italy and UK), institutional investors usually make up the bulk of nonbank holdings. Foreign official sector covers (i) foreign central banks holding other country debt securities as reserve assets and (ii) foreign official lending in the form of bilateral or multilateral official loans

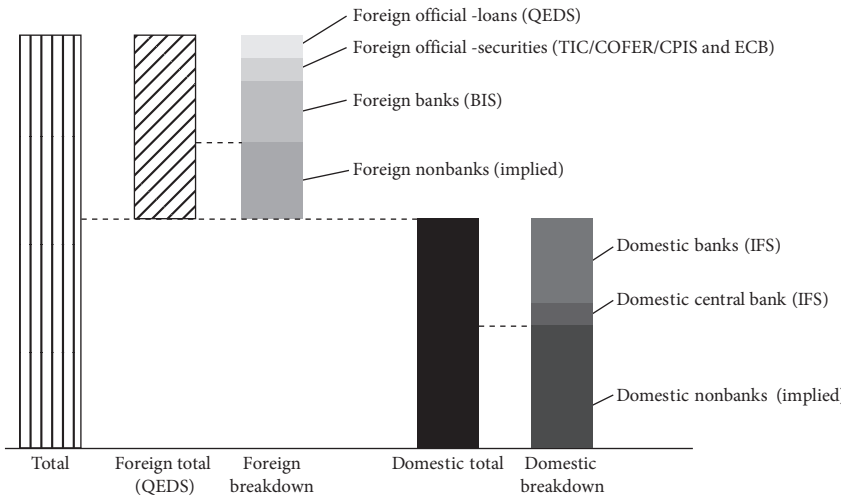

Figure 2.21 Compiling of sovereign investor base estimates—summary of methodology
Source: Arslanalp and Tsuda (2014a).

debt levels and explains (alongside the government's considerable financial assets) why you hear little concern expressed about what is, on the face of it, a very high level of sovereign debt.

The methodology used to compile the investor base holding of sovereign debt are summarized in Figure 2.21. The approach has the following characteristics: First, a common definition of sovereign debt is used—general government gross debt covering currency and deposits; debt securities; and loans. Second, a common estimation methodology is used to ensure cross-country comparability based on harmonized international data sources, such as the BIS, IMF, and World Bank. Third, all data are compiled in face value to track investor transactions as well as holdings. Fourth, foreign investor holdings are estimated separately for the foreign official sector, foreign banks, and foreign nonbanks, in contrast to national data sources that usually classify them under one category ("rest of the world").

The estimates have been published online since 2012 with semi-annual updates[20] and are summarized in Figures 2.22 and 2.23.

In some cases, this approach could be extended to estimate the country of origin of investor holdings. For illustration, Figure 2.24 shows the geographical

[20] See IMF, Sovereign Debt Investor Base for Advanced Economies, available at https://www.imf.org/~/media/Websites/IMF/imported-datasets/external/pubs/ft/wp/2012/Data/_wp12284.ashx and Sovereign Debt Investor Base for Emerging Markets, available at https://www.imf.org/~/media/Websites/IMF/imported-datasets/external/pubs/ft/wp/2014/Data/wp1439.ashx

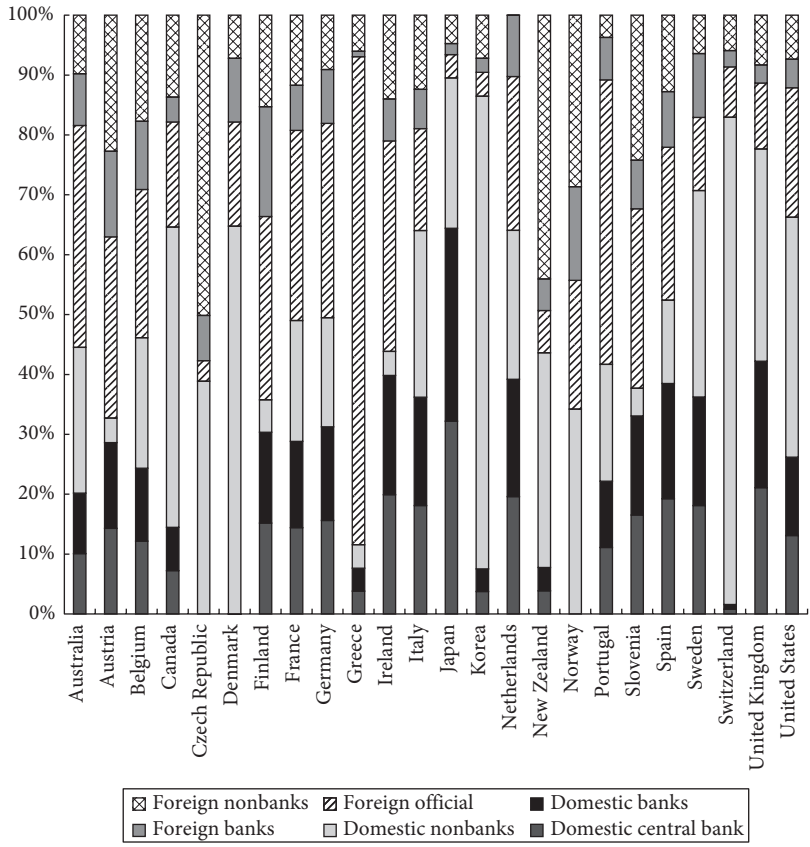

Figure 2.22 Advanced economies: holders of general government debt, end-2017 (in % of total amount outstanding)
Source: Arslanalp and Tsuda (2014a) updated.

decompositions of the investor base of the general government debt of Greece and Japan. Examining investors' country of origin can help assess spillover channels (e.g., euro area holdings of Greek debt), as well as emerging regional linkages (e.g., Chinese investment in Japan).

Finally, having a view of investors across countries is essential for understanding the dynamics of global demand for government debt. Changes in global investor's allocations among countries are important because they can affect many countries all at once. For example, during 2010–13, foreign official holders replaced almost all the foreign private holders of Greece's sovereign debt. In contrast, some of these foreign private investors appear to have shifted to safer assets, such as German bunds, as shown in Figure 2.25 for illustrative purposes.

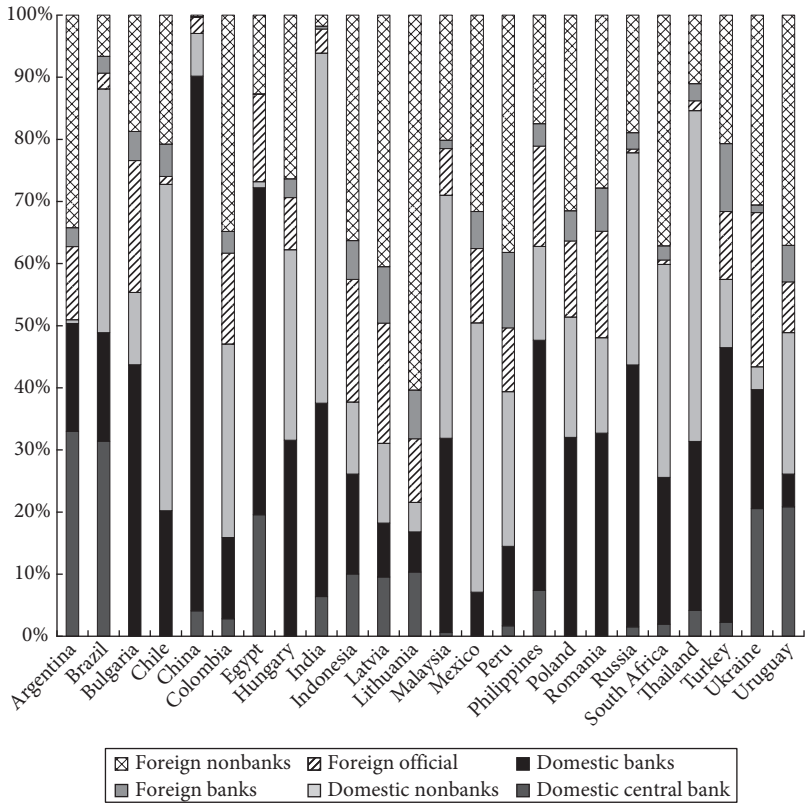

Figure 2.23 Emerging markets: holders of general government debt, end-2017 (in % of total amount outstanding)

Source: Arslanalp and Tsuda (2014b) updated.

5. Conclusions

When Fred Durst erected his debt clock in 1989, it was to raise awareness of what he saw as the United States' rising debt and the burden he thought it would cause future generations. But Mr. Durst's clock, and its imitators in other countries, provide an overly simplistic picture of government debt and fail to capture its many complexities.

Which entities are included in the measure of debt, which instruments are captured and how they are valued are all important pieces of information. Information on risks outside the balance sheet, such as contingent liabilities, should also be taken into account. And additional information on the currency of denomination, maturity, creditor profile, and legal jurisdiction would be needed to complete the picture on government liabilities. To complete the

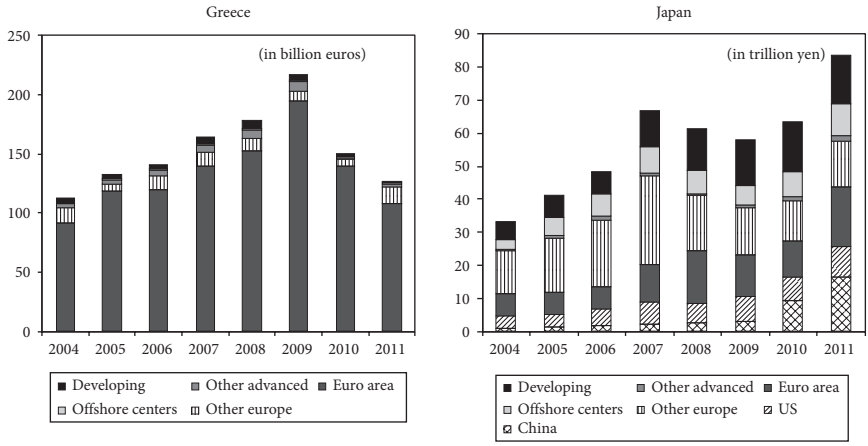

Figure 2.24 Foreign holdings of government debt by country of origin

Notes: Excluding foreign official loans andSMP holdings of foreign central banks. Regional groups are based on country classifications of BIS international banking statistics.

Sources: IMF CPISand authors' calculations.

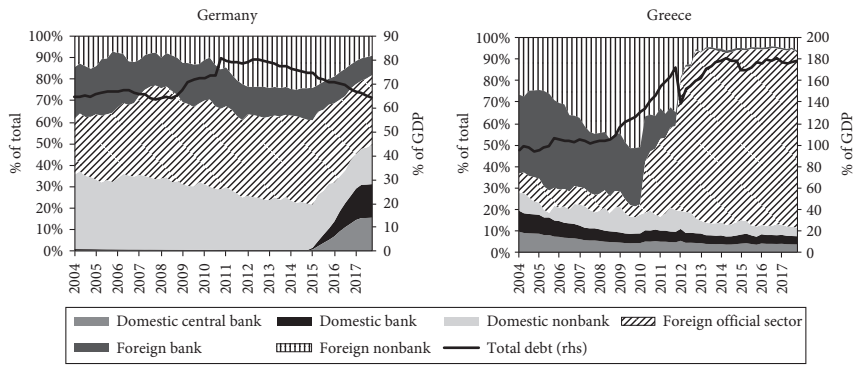

Figure 2.25 Holders of advanced economy general government debt, 2004–17 (components in %; total in % of GDP)

Source: Arslanalp and Tsuda (2014a) updated.

fiscal picture, one would also need to incorporate information on a country's public sector liabilities outside government, in state-owned enterprises, as well as the assets the public sector owns.

A debt clock or headline measure of debt can be seen as analogous to the odometer in a car, measuring the total debt acquired at a certain point in time using a set metric. However, what may really be needed to successfully pilot your macroeconomy may be a suite of different measures of stocks and flows that more closely resembles the cockpit of a jumbo jet. This chapter provides an overview of the reasons why the answer to the question "What

is the national debt of the United States?" can generate answers differing by trillions of dollars and, as with all macroeconomic statistics or accounting data, highlights the importance of metadata to ensure you fully appreciate how to interpret the data.

Annex 1. A country coverage in IMF debt sustainability analysis

The IMF takes a keen interest in the public finances, and the debt, of its member countries. IMF Article IV reports include fiscal tables and Debt Sustainability Analyses (DSA) for its member states, which are presented to the IMF Board, and published online. In addition, the IMF releases databases and publications (such as the World Economic Outlook and Fiscal Monitor) which provide information on government debt.

Despite IMF attempts to codify debt, and provide standard definitions, for example in the 2011 IMF *Public Sector Debt Guide for Compilers and Users* and *the Government Finance Statistics Manual 2014* and earlier editions, debt data used by the Fund for DSA remains a mixture of central government, general government, and other types of coverage, such as the nonfinancial public sector. This variation in which entities are included in debt, although disclosed in metadata, reduces the extent to which debt data in IMF publications and databases is cross-country comparable.

Last DSA Issuance	Total	AE	EM
General government	59	27	32
Central government	31	4	27
Nonfinancial public sector	9	0	9
Consolidated public sector	8	2	6
Other*	5	0	5

Annex 2. Statement of fiscal risks (taken from Everaert et al. 2009)

A statement of fiscal risks can be structured by grouping similar risks: macroeconomic risks (e.g., from growth, terms-of-trade, and exchange and interest rates); contingent obligations (e.g., government guarantees); risks due to the operations of public–private partnerships (PPPs) and state-owned enterprises (SOEs); central government backing of subnational levels of government; risks related to natural disasters; and fluctuations in the value of public sector assets. To avoid moral hazard, implicit risks—including those from the banking system and ongoing litigation against the state—in principle should not be disclosed. However, actions already announced or undertaken should be fully disclosed and discussed.

For each type of relevant risk, the statement could discuss past realization and forward-looking risk estimates. The discussion and quantification of past risks provides background to policies aimed to reduce such risks in the future. For instance, systemic revenue overestimation points to the need for more detailed analysis of the underlying assumptions

and method for estimating revenue, including the economic growth assumption. Similarly, frequent bailouts of PPPs, SOEs, and subnational levels of government may call for strengthening the monitoring and central control of their activities.

Forward-looking risk estimates can draw on existing country practices for disclosing risks, that is, (i) sensitivity analysis to key macroeconomic variables, alternative macroeconomic scenarios, stress tests for fiscal aggregates, or fan charts that illustrate the probability distribution for outcomes; (ii) debt sustainability analyses; (iii) description and quantification of budget exposure to government guarantees through option pricing models, stochastic simulation, or risk ratings approaches; (iv) description of government guarantees in PPP projects, alongside the projects' face value, expected cash flow payments by the government and their net present value; and (v) the nature and scope of ongoing litigation against the state.

In addition, full-fledged general government or public sector accounts and timely audited SOE accounts provide a good source of information for a statement of fiscal risks.

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